

Air Conditioning (A/C) System Recovery, Evacuation and Charging



Refrigerant System Recovery

NOTICE: Use Refrigerant Identification Equipment before recovering any of the vehicle's refrigerant. Failure to do so puts the shop's bulk refrigerant at risk of contamination. If the vehicle's A/C refrigerant is contaminated, refer the customer to the service facility that carried out the last A/C service. If the customer wishes to pay the additional cost, use the A/C recovery equipment that is designated for recovering contaminated A/C refrigerant. All contaminated A/C refrigerant must be disposed of as hazardous waste in accordance with all federal, state and local regulations. For all equipment, follow the equipment manufacturer procedures and instructions.

NOTE: Ford Motor Company recommends the use of R-134a refrigerant management equipment that meets the requirements of the SAE J2788 standard.

1. Verify the refrigerant purity prior to recovery. For additional information, refer to Refrigerant Identification Testing.
 - 198-RI2012yfp A/C Refrigerant Analyzer for use with R134a and R1234yf vehicles
Use the General Equipment: Air Conditioning Service Unit
2. Connect a R-134a Air Conditioning Service Unit to the low- and high-pressure service gauge port valves following the operating instructions provided by the equipment manufacturer.
 - 265-37887 Ritchie R134A A/C Refrigerant Hybrid Management System
Use the General Equipment: Air Conditioning Service Unit
3. **NOTE:** *Never attempt to recover from only the low side service gauge port valve (solenoid valves in the system do not allow for full recovery).*

Recover the refrigerant from the system following the operating instructions provided by the equipment manufacturer. Note the amount of oil removed during the refrigerant recovery (if any). Add that same amount back into the system once repairs are complete.

4. Once the R-134a Air Conditioning Service Unit has recovered the refrigerant, switch OFF the power supply.
5. Allow the system to set for about 2 minutes, and observe the system vacuum reading. If the vacuum is not lost, disconnect the recovery equipment.
6. If the system loses vacuum, repeat Steps 3 through 5 until the vacuum level remains stable for 2 minutes.
7. Carry out the required repairs.

Refrigerant System Evacuation

1. **NOTICE:** Use Refrigerant Identification Equipment before recovering any of the vehicle's refrigerant. Failure to do so puts the shop's bulk refrigerant at risk of contamination. If the vehicle's A/C refrigerant is contaminated, refer the customer to the service facility that carried out the last A/C service. If the customer wishes to pay the additional cost, use the A/C recovery equipment that is designated for recovering contaminated A/C refrigerant. All contaminated A/C refrigerant must be disposed of as hazardous waste in accordance with all federal, state and local regulations. For all equipment, follow the equipment manufacturer procedures and instructions.

- 198-RI2012yfp A/C Refrigerant Analyzer for use with R134a and R1234yf vehicles
Use the General Equipment: Refrigerant Identification Equipment

2. Connect the Air Conditioning Service Unit to the low- and high-pressure service gauge port valves following the equipment instructions.
 - 265-37887 Ritchie R134A A/C Refrigerant Hybrid Management System
Use the General Equipment: Air Conditioning Service Unit

NOTE: *Never attempt to recover from only the low side service gauge port valve (solenoid valves in the system do not allow for full recovery).*

3. Evacuate the system until the low-pressure gauge reads at least 99.4 kPa (29.5 in-Hg) of vacuum and as close to 101.1 kPa (30 in-Hg) as possible. Continue to operate the vacuum pump for a minimum of 45 minutes.
4. Turn OFF the vacuum pump. Observe the low-pressure gauge for 10 minutes to make sure that the system vacuum is held.

Refrigerant System Charging

1. **NOTICE:** Use Refrigerant Identification Equipment before recovering any of the vehicle's refrigerant. Failure to do so puts the shop's bulk refrigerant at risk of contamination. If the vehicle's A/C refrigerant is contaminated, refer the customer to the service facility that carried out the last A/C service. If the customer wishes to pay the additional cost, use the A/C recovery equipment that is designated for recovering contaminated A/C refrigerant. All contaminated A/C refrigerant must be disposed of as hazardous waste in accordance with all federal, state and local regulations. For all equipment, follow the equipment manufacturer procedures and instructions.
 - 198-RI2012yfp A/C Refrigerant Analyzer for use with R134a and R1234yf vehicles
Use the General Equipment: Refrigerant Identification Equipment
2. Lubricate the refrigerant system with the correct amount of clean PAG oil. For additional information, refer to Refrigerant Oil Adding.
3. Connect the Air Conditioning Service Unit to the low- and high-pressure service gauge port valves following the equipment instructions.
 - 265-37887 Ritchie R134A A/C Refrigerant Hybrid Management System
Use the General Equipment: Air Conditioning Service Unit
4. Set the refrigerant charge amount and charge the refrigerant system following the equipment instructions.